



IEEE Standard for Ethernet

Amendment 1: Power over Data Lines of Single Pair Ethernet

IEEE Computer Society

Developed by the
LAN/MAN Standards Committee

IEEE Std 802.3dd™-2022
(Amendment to IEEE Std 802.3™-2022)

IEEE Standard for Ethernet

Amendment 1: Power over Data Lines of Single Pair Ethernet

LAN/MAN Standards Committee
of the
IEEE Computer Society

Approved 16 June 2022
IEEE SA Standards Board

Abstract: This amendment to IEEE Std 802.3-2022 makes editorial and technical corrections, refinements, and clarifications to Clause 104 Power over Data Lines of Single Pair Ethernet, and related portions of the standard. No new to features are added by the amendment.

Keywords: 10BASE-T1L, 10BASE-T1S, 100BASE-T1, 1000BASE-T1, 2.5GBASE-T1, 5GBASE-T1, 10GBASE-T1, copper, Ethernet, IEEE 802.3™, IEEE 802.3dd™, PoDL

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Introduction

This introduction is not part of IEEE Std 802.3dd-2022, IEEE Standard for Ethernet—Amendment 1: Power over Data Lines of Single Pair Ethernet.

IEEE Std 802.3™ was first published in 1985. Since the initial publication, many projects have added functionality or provided maintenance updates to the specifications and text included in the standard. Each IEEE 802.3 project/amendment is identified with a suffix (e.g., IEEE Std 802.3ba™-2010).

The half duplex Media Access Control (MAC) protocol specified in IEEE Std 802.3-1985 is Carrier Sense Multiple Access with Collision Detection (CSMA/CD). This MAC protocol was key to the experimental Ethernet developed at Xerox Palo Alto Research Center, which had a 2.94 Mb/s data rate. Ethernet at 10 Mb/s was jointly released as a public specification by Digital Equipment Corporation (DEC), Intel and Xerox in 1980. “Local Area Networks: Carrier sense multiple access with collision detection (CSMA/CD) access method and physical layer specifications” was approved as an IEEE standard by the IEEE Standards Board in 1983 and subsequently published in 1985 as IEEE Std 802.3-1985. Since 1985, new media options, new speeds of operation, and new capabilities have been added to IEEE Std 802.3. A full duplex MAC protocol and the ability to use an EtherType to specify the MAC client protocol were added in 1997. The title was changed to Standard for Ethernet with the 2012 Revision.

Some of the major additions to IEEE Std 802.3 are identified in the marketplace with their project number. This is most common for projects adding higher speeds of operation or new protocols. For example, IEEE Std 802.3u™ added 100 Mb/s operation (also called Fast Ethernet), IEEE Std 802.3z™ added 1000 Mb/s operation (also called Gigabit Ethernet), IEEE Std 802.3ae™ added 10 Gb/s operation (also called 10 Gigabit Ethernet), IEEE Std 802.3ah™ specified access network Ethernet (also called Ethernet in the First Mile) and IEEE Std 802.3ba added 40 Gb/s operation (also called 40 Gigabit Ethernet) and 100 Gb/s operation (also called 100 Gigabit Ethernet). These major additions are all now included in and are superseded by IEEE Std 802.3-2022 and are not maintained as separate documents.

At the date of this IEEE Std 802.3dd-2022 publication, IEEE Std 802.3 was composed of the following documents:

IEEE Std 802.3-2022

Section One—Includes Clause 1 through Clause 20 and Annex A through Annex K and Annex 4A. Section One includes the specifications for 10 Mb/s operation and the MAC, frame formats and service interfaces used for all speeds of operation.

Section Two—Includes Clause 21 through Clause 33 and Annex 22A through Annex 33A. Section Two includes management attributes for multiple protocols and speed of operation as well as specifications for providing power over twisted pair cabling for multiple operational speeds. It also includes general information on 100 Mb/s operation as well as most of the 100 Mb/s Physical Layer specifications.

Section Three—Includes Clause 34 through Clause 43 and Annex 36A through Annex 43C. Section Three includes general information on 1000 Mb/s operation as well as most of the 1000 Mb/s Physical Layer specifications.

Section Four—Includes Clause 44 through Clause 55 and Annex 44A through Annex 55B. Section Four includes general information on 10 Gb/s operation as well as most of the 10 Gb/s Physical Layer specifications.

Section Five—Includes Clause 56 through Clause 77 and Annex 57A through Annex 76A. Clause 56 through Clause 67 and Clause 75 through Clause 77, as well as associated annexes, specify subscriber access and other Physical Layers and sublayers for operation from 512 kb/s to 10 Gb/s, and defines services and protocol elements that enable the exchange of IEEE Std 802.3 format frames between stations in a subscriber access network. Clause 68 specifies a 10 Gb/s Physical Layer specification. Clause 69 through Clause 74 and associated annexes specify Ethernet operation over electrical backplanes at speeds of 1000 Mb/s and 10 Gb/s.

Section Six—Includes Clause 78 through Clause 95 and Annex 83A through Annex 93C. Clause 78 specifies Energy-Efficient Ethernet. Clause 79 specifies IEEE 802.3 Organizationally Specific Link Layer Discovery Protocol (LLDP) type, length, and value (TLV) information elements. Clause 80 through Clause 95 and associated annexes include general information on 40 Gb/s and 100 Gb/s operation as well as 40 Gb/s and 100 Gb/s Physical Layer specifications. Clause 90 specifies Ethernet support for time synchronization protocols.

Section Seven—Includes Clause 96 through Clause 115 and Annex 97A through Annex 115A. Clause 96 through Clause 98, Clause 104, and associated annexes, specify Physical Layers and optional features for 100 Mb/s and 1000 Mb/s operation over a single twisted pair. Clause 100 through Clause 103, as well as associated annexes, specify Physical Layers for the operation of the EPON protocol over coaxial distribution networks. Clause 105 through Clause 114 and associated annexes include general information on 25 Gb/s operation as well as 25 Gb/s Physical Layer specifications. Clause 99 specifies a MAC merge sublayer for the interspersing of express traffic. Clause 115 and its associated annex specify a Physical Layer for 1000 Mb/s operation over plastic optical fiber.

Section Eight—Includes Clause 116 through Clause 140 and Annex 119A through Annex 136D. Clause 116 through Clause 124 and associated annexes include general information on 200 Gb/s and 400 Gb/s operation as well as 200 Gb/s and 400 Gb/s Physical Layer specifications. Clause 125 includes general information on 2.5 Gb/s and 5 Gb/s operation. Clause 126 through Clause 130 and associated annexes include 2.5 Gb/s and 5 Gb/s Physical Layer specifications. Clause 131 provides general information on 50 Gb/s operation. Clause 132 through Clause 140 and associated annexes include 50 Gb/s Physical Layer specifications and additional 100 Gb/s, 200 Gb/s, and 400 Gb/s Physical Layer specifications.

Section Nine—Includes Clause 141 through Clause 160 and Annex 142A through Annex 154A. Clause 141 through Clause 144 and associated annexes specify symmetric and asymmetric operation of Ethernet passive optical networks over multiple 25 Gb/s channels. Clause 145 and associated annexes specify increased power delivery using all four pairs in the structured wiring plant. Clause 146 through Clause 149 and associated annexes specify Physical Layers for 10 Mb/s, 2.5 Gb/s, 5 Gb/s, and 10 Gb/s operation over a single balanced pair of conductors. Clause 150 and Clause 151 include additional 400 Gb/s Physical Layer specifications. Clause 153 and Clause 154 specify 100 Gb/s operation over DWDM channels. Clause 157 through Clause 160 include 10 Gb/s, 25 Gb/s, and 50 Gb/s bidirectional Physical Layer specifications.

IEEE Std 802.3dd™-2022

Amendment 1—This amendment includes editorial and technical corrections, refinements, and clarifications to Clause 104, Power over Data Lines of Single Pair Ethernet, and related portions of the standard.

Two companion documents exist, IEEE Std 802.3.1 and IEEE Std 802.3.2. IEEE Std 802.3.1 describes Ethernet management information base (MIB) modules for use with the Simple Network Management Protocol (SNMP). IEEE Std 802.3.2 describes YANG data models for Ethernet. IEEE Std 802.3.1 and

IEEE Std 802.3.2 are updated to add management capability for enhancements to IEEE Std 802.3 after approval of those enhancements.

IEEE Std 802.3 will continue to evolve. New Ethernet capabilities are anticipated to be added within the next few years as amendments to this standard.

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IEEE Standard for Ethernet

Amendment 1: Power over Data Lines of Single Pair Ethernet

(This amendment is based on IEEE Std 802.3™-2022.)

NOTE—The editing instructions contained in this amendment define how to merge the material contained therein into the existing base standard and its amendments to form the comprehensive standard.

The editing instructions are shown in ***bold italic***. Four editing instructions are used: change, delete, insert, and replace. ***Change*** is used to make corrections in existing text or tables. The editing instruction specifies the location of the change and describes what is being changed by using ~~strike through~~ (to remove old material) and underscore (to add new material). ***Delete*** removes existing material. ***Insert*** adds new material without disturbing the existing material. Deletions and insertions may require renumbering. If so, renumbering instructions are given in the editing instruction. ***Replace*** is used to make changes in figures or equations by removing the existing figure or equation and replacing it with a new one. Editing instructions, change markings, and this NOTE will not be carried over into future editions because the changes will be incorporated into the base standard.

Cross references that refer to clauses, tables, equations, or figures not covered by this amendment are highlighted in **green**.⁶

⁶Notes in text, tables, and figures are given for information only and do not contain requirements needed to implement the standard.

104. Power over Data Lines (PoDL) of Single-Pair Ethernet

Change the title of 104.3 as shown:

104.3 ~~Class power requirements~~ System parameters

Insert the following two new paragraphs at the end of 104.3:

V_{PSE} is the voltage at the PSE PI. V_{PSE} is measured between any positive conductor and any negative conductor at the PI of the PSE.

V_{PD} is the voltage at the PD PI. V_{PD} is measured between any positive conductor and any negative conductor at the PI of the PD.

104.5 Powered Device (PD)

104.5.4 PD State Diagram

104.5.4.3 Variables

Change the definition of the variables `sccp_reset_pulse` and `wakeup` as shown (unchanged variables not shown):

`sccp_reset_pulse`

TRUE: ~~during detection,~~ a SCCP reset pulse per Figure 104–10 as described in 104.7.1.1 has been received by the PD.

FALSE: ~~during detection,~~ a SCCP reset pulse has not been received by the PD.

`wakeup`

An implementation specific variable enabling the PD to request wakeup.

TRUE: the PD requires the full operating voltage at the PI.

FALSE: the PD is ready to go to sleep or is otherwise not requesting full operating voltage.

104.5.4.6 State diagram

Change Figure 104-8, changing state *DO_CLASSIFICATION*, and adding exit branch *A* to *PD_SLEEP*, entering *DO_CLASSIFICATION*, as shown:

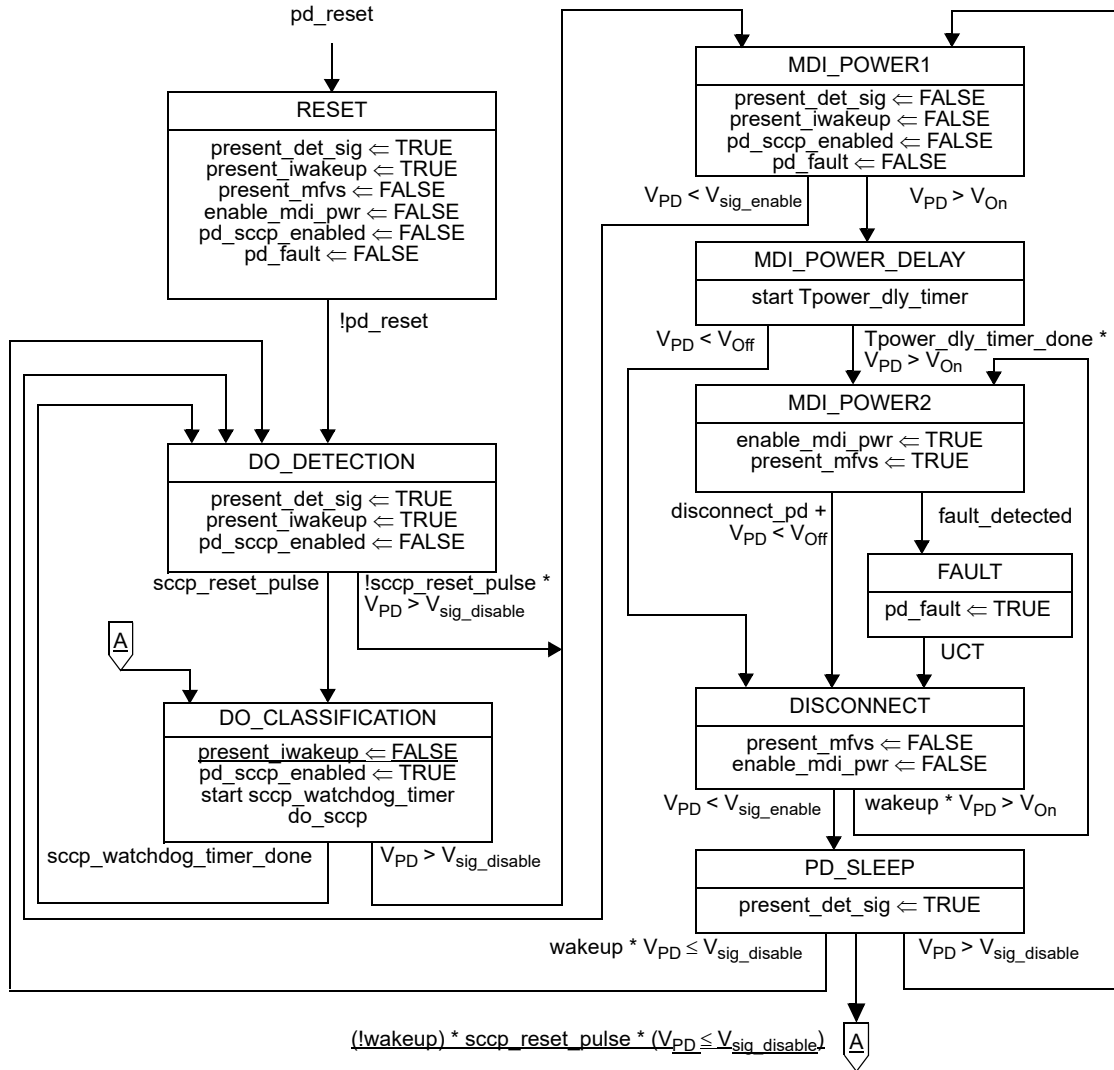


Figure 104–8—PD state diagram

104.5.5 PD signature

Change Table 104-9 as follows:

Table 104-9—Valid PD detection signature characteristics, measured at PD PI

Parameter	Conditions	Min	Max	Unit
V_{good}	$7 \text{ mA} < I_{\text{PD}} < 17 \text{ mA}$, PD exiting RESET state	4.05	4.55	V
$I_{\text{signature_limit}}$	$V_{\text{PD}} < V_{\text{sig_disable_max}}$		24	mA
$V_{\text{sig_disable}}$ Classes 0 to 9	V_{PD} rising	4.6	5.75	V
$V_{\text{sig_disable}}$ Classes 10 to 15	V_{PD} rising	6.0	7.5	V
$V_{\text{sig_enable}}$	V_{PD} falling	3.6	4.3	V

104.5.7 PD power

Change Table 104-11 items 6b and 15, as follows (unchanged rows not shown):

Table 104-11—PD power supply limits

Item	Parameter	Symbol	Unit	Min	Max	PD Type	Additional information
...							
6b	Input capacitance during DO_CLASSIFICATION state	C _{IN_Class}	μF	—	0.2	A, B, C, D	All classes <u>Applies during t_R and t_F only</u>
				—	0.4 <u>0.8</u>	E	
...							
15	SCCP watchdog timeout	T _{SCCP_watchdog}	ms	150	200	A, B, C, D	See 104.5.6
				1000 <u>1300</u>	1300 <u>2000</u>	E	

104.5.7.1 PD discharge

Change text of 104.5.7.1 as shown:

At a delay of $T_{\text{OFF_max}}$ (see Table 104-7) after disconnection from the PSE, a PD shall not source greater than $410 \mu\text{J}$ out of its PI until V_{PD} drops below $V_{\text{Sleep_PD_max}}$. When there is no PSE or the PSE is not sourcing power, the PD might present a voltage on the (unpowered) pair. This can cause a current to flow from the PD. In order to constrain this current, the voltage across a $5 \text{ k}\Omega$ resistor connected across the PD PI shall not exceed V_{PUP} as defined in Table 104-12, after a delay of $T_{\text{OFF_max}}$ (see Table 104-7) after the removal of PSE power from the PD PI.

104.5.7.3 Input current

Change the first paragraph of 104.5.7.3 as shown:

During operation in the ~~DISCONNECT~~ and PD_SLEEP states, the PD shall not draw current in excess of $I_{\text{Sleep_PD}}$ as specified in Table 104–11.

104.6 Additional electrical specifications

104.6.1 Isolation

Insert the following sentence at the end of the first paragraph of 104.6.1:

Compliance with requirements of 104.8.1 may require greater isolation, particularly in applications outside of automotive or other contained systems.

104.6.2 Fault tolerance

Change the first paragraph in 104.6.2 as follows:

The PI for Type A, Type B, and Type C PSEs and PDs shall meet the fault tolerance requirements as specified in 96.8.3. The PI for Type E ~~PSEs and~~ PDs shall meet the fault tolerance requirements as specified in 146.8.6.

104.7 Serial communication classification protocol (SCCP)

104.7.1 SCCP signaling

104.7.1.1 Initialization procedure—reset and presence pulses

Replace Figure 104-10 to remove t_{CHRG} and V_{CHRG} as follows:

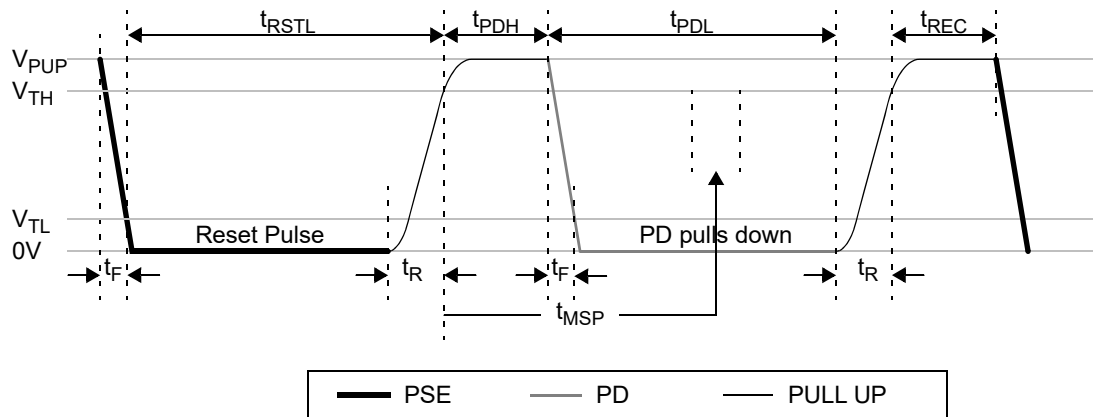


Figure 104–10—Reset command timing diagram

104.7.1.2 Write time slots

Replace Figure 104-11 to remove t_{CHRG} and V_{CHRG} as follows:

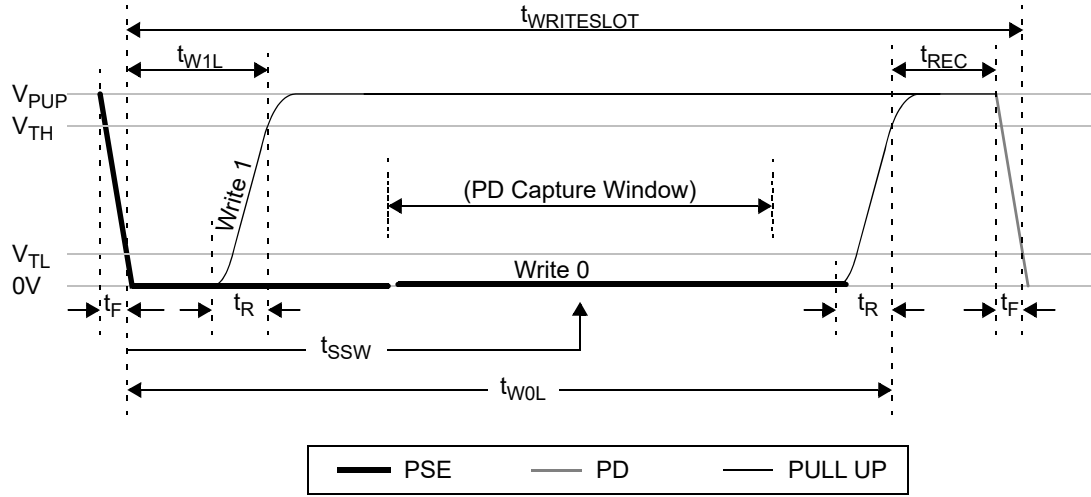


Figure 104-11—Write 0/1 slot timing diagram

104.7.1.3 Read time slots

Replace Figure 104-12 to remove t_{CHRG} and V_{CHRG} as follows:

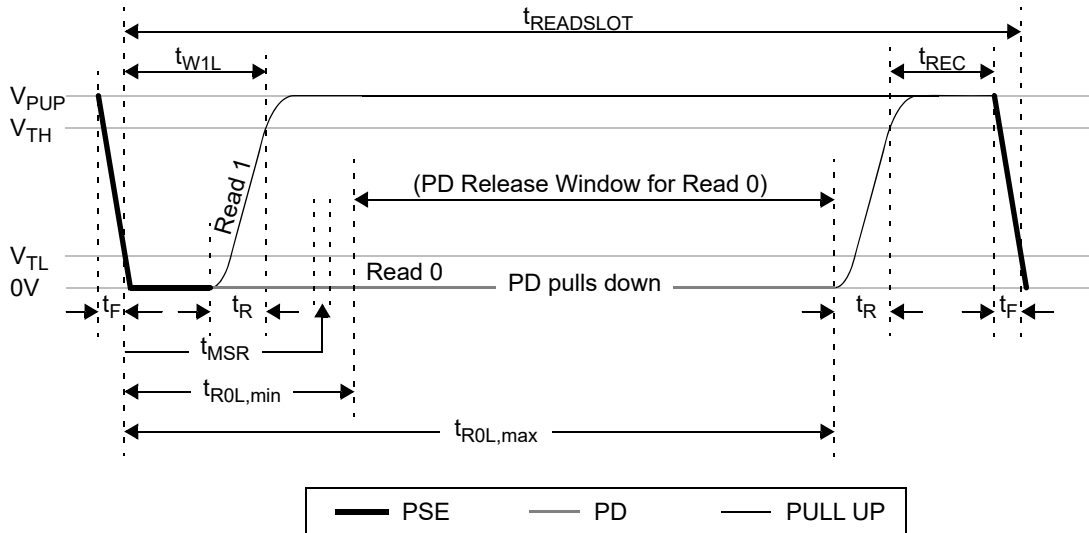


Figure 104-12—Read 0/1 slot timing diagram

Change Table 104-12 as follows, editing rows 6a, 6b, 7, 8, 9, 11, 15, 16, and 18, and removing rows 20 and 21, unchanged rows not shown:

Table 104–12—SCCP electrical requirements

Item	Parameter	Symbol	Unit	Min	Max	PSE/ PD Type	Additional information
...							
6a	Write Time Slot	t _{WRITESLOT}	ms	2.7	3.3	A, B, C, D	
				—	3.852 78	E	
6b	Read Time Slot	t _{READSLOT}	ms	2.7	3.3	A, B, C, D	
				—	53.83	E	
7	Recovery Time	t _{REC}	ms	0.27	0.33	A, A, B, C, D	
				0.5	—	E	
8	Write 0 Low Time	t _{W0L}	ms	1.8	2.2	A, A, B, C, D	
				1.8	2.6	E	
9	Write 1 Low Time	t _{W1L}	ms	0.08	0.25	A, B, C, D	
				0.09	0.644	E	
...							
11	PSE Sample Read Time	t _{MSR}	ms	0.27	0.33	A, B, C, D	
				0.9	1.34	E	
...							
15	Presence-Detect Low Time	t _{PDLOW}	ms	2.5	7.5	A, B, C, D	
				2.8	5.2	E	
				21	394	E	PDs that support link segment resistance measurement
16	PSE Sample Presence Time	t _{MSP}	ms	1.8	2.2	A, A, B, C, D	
				1.8	2.4	E	
...							

Table 104–12—SCCP electrical requirements (*continued*)

Item	Parameter	Symbol	Unit	Min	Max	PSE/ PD Type	Additional information
18	Fall-Time	t_F	ms	0.025	0.1	A, B, C, D	
				0.025	0.25	E	
...							
20	PD-reservoir capacitor recharge voltage	V_{CHRG}	V	$0.9 \times V_{PUPmin}$	—	E	
21	PD-reservoir capacitor recharge time	t_{CHRG}	ms	0.2	—	E	

104.7.2 Serial communication classification protocols

Change first paragraph of 104.7.2 as shown:

All data and commands shall be transmitted least significant bit first using SCCP. The PSE initiates all transactions. The PD shall return all 1s in the payload and CRC8 fields for any unsupported command.

104.7.2.6 Read_VOLT_INFO command [0xBB]

Insert new final sentence in first paragraph of 104.7.2.6 as follows:

The voltage measurement returned by the Read_VOLT_INFO command is an 8-bit unsigned value with each least-significant bit equal to 10 mV. The value returned may be less than the actual PD PI voltage and may be any value up to 20 mV greater than the actual voltage at the PD PI.

Change Table 104-14, Description for b[7:0] as shown.:

Table 104–14—VOLT_INFO Register Table

Bit(s)	Name	Description	R/W ^a
b[15:8]	Reserved	Value Always 0	RO
b[7:0]	Voltage at PD PI during Presence Pulse	± 20 mV tolerance, 10 mV per LSB	RO

^aRO = Read only

104.9 Protocol implementation conformance statement (PICS) proforma for Clause 104, Power over Data Lines (PoDL) of Single-Pair Ethernet⁶

104.9.4 PICS proforma tables for Clause 104, Power over Data Lines (PoDL) of Single-Pair Ethernet

104.9.4.3 Powered Device (PD)

Change PICS items PD11 and PD17 as shown (unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
PD11	PD discharge	104.5.7.1	At a delay of T_{OFF_max} after disconnection from the PSE, a PD shall not source greater than 440 μA out of its PI until V_{PD} drops below $V_{Sleep_PD_max}$. The voltage across a 5 kΩ resistor connected across the PD PI shall not exceed V_{PUP} as defined in Table 104-12, after a delay of T_{OFF_max} (see Table 104-7) after the removal of PSE power from the PD PI.	M	Yes []
PD17	Input current while in DISCONNECT and PD_SLEEP states	104.5.7.3	Current drawn is not in excess of I_{Sleep_PD} as specified in Table 104-11	M	Yes []

104.9.4.4 Common Electrical

Change PICS item COMEL2 in 104.9.4.4 (unchanged items not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
COMEL2	Type E PSE and PD fault tolerance	104.6.2	The PI shall meet the fault tolerance requirements as specified in 146.8.6	PSETE:M PDTE:M	Yes [] N/A []

104.9.4.7 SCCP

Insert new PICS item SCCP21a between PICS items SCCP21 and SCCP22 as shown (unchanged items not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
SCCP21a	Unsupported SCCP command	104.7.2	PD returns all 1s in the data and CRC8 fields for any unsupported command	SCCP:M	Yes [] N/A []

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146. Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA) sublayer and baseband medium, type 10BASE-T1L

146.5 PMA electrical specifications

146.5.4 Transmitter electrical specifications

146.5.4.2 Transmitter output droop

Change 146.5.4.2 as shown:

With the transmitter in test mode 2 and using the transmitter test fixture shown in [Figure 146-20](#),

When a Clause 104 Type E PSE or PD PI is not encompassed within the MDI, the magnitude of both the positive and negative droop shall be less than 10% measured with respect to an initial value at 133.3 ns after the zero crossing and a final value at 800 ns after the zero crossing.

When a Clause 104 Type E PSE or PD PI is encompassed within the MDI, the magnitude of both the positive and negative droop shall be less than 25% measured with respect to an initial value at 400 ns after the zero crossing and a final value at 1066.7 ns after the zero crossing. Implementers should consider transmitter amplitude limitations when appropriate to the application such as those applications addressed in [Annex 146A](#). This specification is at the MDI with the MDI loaded as specified in 146.5.4. Addition of power coupling networks and measurement noise can increase the apparent droop. For example, a power coupling network might add 3% and noise might further increase droop by 2%, adding up to 5%. Limiting additional apparent droop due to test setup is strongly encouraged and should be corrected for according to the measurement configuration being used.

146.8 MDI specification

146.8.3 MDI return loss

Change 146.8.3, editing Equation (146-17) and including additional Equation (146-17a), as shown:

When a Clause 104 Type E PSE or PD PI is not encompassed within the MDI, the MDI return loss (RL) shall meet or exceed Equation (146-17) for all frequencies from 100 kHz to 20 MHz (with $100\ \Omega \pm 0.1\%$ reference impedance) at all times when the PHY is transmitting data or idle symbols.

$$\text{Return Loss } (f) \geq \left\{ \begin{array}{ll} 20 - 18 \times \log_{10} \left(\frac{0.2}{f} \right) \text{ dB} & 0.1 \leq f < 0.2 \text{ MHz} \\ 20 \text{ dB} & 0.2 \leq f \leq 1 \text{ MHz} \\ 20 - 16.7 \times \log_{10} (f) \text{ dB} & 1 < f \leq 10 \text{ MHz} \\ 3.3 - 7.6 \times \log_{10} \left(\frac{f}{10} \right) \text{ dB} & 10 < f \leq 20 \text{ MHz} \end{array} \right\} \quad (146-17)$$

where f is the frequency in MHz.

When a Clause 104 Type E PSE or PD PI is encompassed within the MDI, the MDI return loss (RL) shall meet or exceed Equation (146-17a) for all frequencies from 100 kHz to 20 MHz (with $100\ \Omega \pm 0.1\%$ reference impedance) at all times when the PHY is transmitting data or idle symbols.

$$\text{Return Loss } (f) \geq \left\{ \begin{array}{ll} 20 - 21.5 \times \log_{10} \left(\frac{0.5}{f} \right) \text{ dB} & 0.1 \leq f < 0.5 \\ 20 \text{ dB} & 0.5 \leq f \leq 1 \\ 20 - 16.7 \times \log_{10} (f) \text{ dB} & 1 < f \leq 10 \\ 3.3 - 7.6 \times \log_{10} \left(\frac{f}{10} \right) \text{ dB} & 10 < f \leq 20 \end{array} \right\} \quad (146-17a)$$

where f is the frequency in MHz.

146.8.5 MDI DC power voltage tolerance

Change the first paragraph of 146.8.5 as shown:

The DTE, with the exception of MDIs encompassing Clause 104 PSEs, shall withstand without damage the application of any voltages between 0 V dc and 60 V dc with the source current limited to 2000 mA, applied across BI_DA+ and BI_DA–, in either polarity, under all operating conditions, for an indefinite period of time. This requirement ensures that all devices tolerate DC powering voltages, such as those in Clause 104, even if the device does not require power.

146.8.6 MDI fault tolerance

Change the first paragraph of 146.8.6 as shown:

The wire pair of the MDI, with the exception of MDIs encompassing Clause 104 PSEs, shall withstand without damage the application of short circuits of any wire to the other wire of the same pair or ground potential, as per [Table 146-9](#), under all operating conditions, for an indefinite period of time with the source current limited to 2000 mA where applicable due to applied voltage. Normal operation shall resume after the short circuit(s) is/are removed. Clause 104 PSE fault tolerance requirements are defined in 104.6.2.

146.11 Protocol implementation conformance statement (PICS) proforma for Clause 146, Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA) sublayer and baseband medium, type 10BASE-T1L⁷

146.11.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 146, Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA) sublayer and baseband medium, type 10BASE-T1L, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in [Clause 21](#).

146.11.2 Identification

146.11.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

146.11.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3dd-2022, Clause 146, Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA) sublayer and baseband medium, type 10BASE-T1L
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21 ; the answer Yes means that the implementation does not conform to IEEE Std 802.3dd-2022.)	

Date of Statement	
-------------------	--

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146.11.3 Major capabilities/options

Insert two new rows at the end of table in 146.11.3 (unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
*PPSE	MDI encompasses a Clause 104 PSE	146.5.4.2, 146.8.3, 146.8.5 and 146.8.6		O	Yes [] No []
*PPD	MDI encompasses a Clause 104 PD	146.5.4.2 and 146.8.3		O	Yes [] No []

146.11.4 PICS proforma tables for Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA) sublayer and baseband medium, type 10BASE-T1L

146.11.4.5 MDI specifications

Change PICS entries for MDI2 (splitting to create new row MDI2a), MDI4, and MDI5 as shown (unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
MDI2	MDI Return Loss (when a Clause 104 PI is not encompassed)	146.8.3	Meets or exceeds Equation (146–17)	!PPSE* !PPD:M	Yes [] N/A []
<u>MDI2a</u>	<u>MDI Return Loss (when a Clause 104 PI is encompassed)</u>	<u>146.8.3</u>	<u>Meets or exceeds Equation (146–17a)</u>	<u>(PPSE or PPD):M</u>	<u>Yes []</u> <u>N/A []</u>
MDI4	MDI line powering voltage tolerance	146.8.5	Up to 60 V DC with the source current limited to 2000 mA	!PPSE:M	Yes [] N/A []
MDI5	MDI fault tolerance	146.8.6	Withstand without damage the application of a short circuit of any wire to the other wire of the same pair or ground potential, operation resumes after removing the short(s)	!PPSE:M	Yes [] N/A []

147. Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA) sublayer and baseband medium, type 10BASE-T1S

147.9 MDI specification

147.9.3 MDI line powering voltage tolerance

Change the first paragraph of 147.9.3 as shown:

The DTE, with the exception of MDIs encompassing Clause 104 PSEs, shall withstand without damage the application of any voltages between 0 V dc and 60 V dc with the source current limited to 2000 mA, applied across BI_DA+ and BI_DA-, in either polarity, under all operating conditions, for an indefinite period of time.

147.9.4 MDI fault tolerance

Change the first paragraph of 147.9.4 as shown:

The wire pair of the MDI, with the exception of MDIs encompassing Clause 104 PSEs, shall withstand without damage the application of short circuits of any wire to the other wire of the same pair or ground potential, as per [Table 147-5](#), under all operating conditions, indefinitely with the source current limited to 2000 mA where applicable due to applied voltage. Normal operation shall resume after all short circuits have been removed. Clause 104 PSE fault tolerance requirements are defined in 104.6.2.

147.12 Protocol implementation conformance statement (PICS) proforma for Clause 147, Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA) sublayer and baseband medium, type 10BASE-T1S⁸

147.12.3 Major capabilities/options

Insert one new row at the end of table in 147.12.3 (unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
*PPSE	MDI encompasses a Clause 104 PSE	147.9.3 and 147.9.4		O	Yes [] No []

147.12.4 PICS proforma tables for Physical Medium Attachment (PMA) sublayer and baseband medium, type 10BASE-T1S

147.12.4.9 MDI specifications

Change PICS entries for MDI3 and MDI4 as shown (unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
MDI3	MDI line powering voltage tolerance	147.9.3	Up to 60 V dc with the source current limited to 2000 mA	<u>!PPSE</u> :M	Yes [] <u>N/A</u> []
MDI4	MDI fault tolerance	147.9.4	Withstand without damage the application of a short circuit of any wire to the other wire of the same pair or ground potential. Normal operation resumes after all short circuits are removed.	<u>!PPSE</u> :M	Yes [] <u>N/A</u> []

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